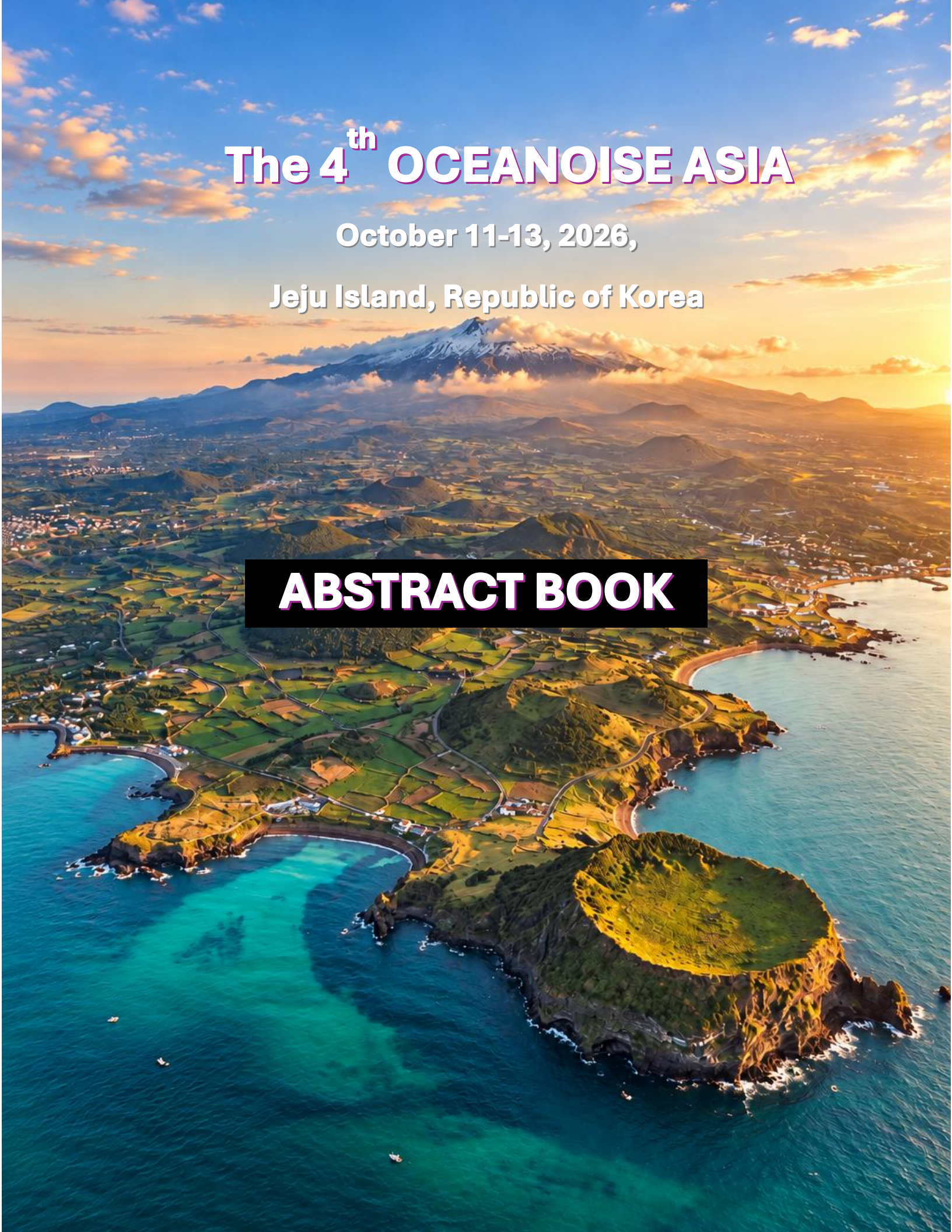


The 4th OCEANOISE ASIA

October 11-13, 2026,

Jeju Island, Republic of Korea

ABSTRACT BOOK

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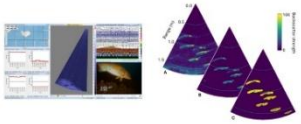
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The 4th Oceanoise Asia (Oceanoise Asia 2026)

October 11-13, 2026, Jeju Island, Republic of Korea

Website: https://www.oceanoiseasia2026.com

We are living in the Anthropocene, where rapid climate change and expanding human activities are transforming the ocean environment at an unprecedented pace. Rising sea temperatures, ocean acidification, and increasing anthropogenic noise are reshaping the acoustic habitats that marine mammals depend on for communication, navigation, and survival. Jeju Island, where surrounding waters are warming substantially faster than the global average, represents a critical location for understanding these emerging environmental and acoustic challenges. Oceanoise Asia 2026 brings together scientists, engineers, policymakers, and conservationists to strengthen regional and global collaboration in understanding and mitigating ocean noise. This October in Jeju, we invite you to join us in shaping a quieter, healthier, and more resilient future for our oceans.

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CONFERENCE SCHEDULE

Sunday (October 11)	Program	Notes
18:00-20:00	Committee gathering	For committee members please check invitation email
Monday (October 12)	**Program**	**Notes**
08:00-09:00	Registration / Morning coffee	Visit registration desk in the Ara Conventional Hall
09:00-09:10	Opening	Opening remarks
09:10-10:00	**Invited talks**	25 minutes + roundtable * 2 talks
10:00-11:00	**Marine bioacoustics (session)**	15 minutes * 4 talks
11:00-11:20	**Marine bioacoustics (discussion)**	20 minutes
11:20-13:00	Lunch	Cafeteria
13:00-14:30	**Sonar and signal processing (session)**	15 minutes * 6 talks
14:30-14:50	**Sonar and signal processing (discussion)**	20 minutes
14:50-15:20	Coffee break	30 minutes
15:20-16:35	**Noise and effects (session)**	15 minutes * 5 talks
16:35-16:55	**Noise and effects (discussion)**	20 minutes
16:55-17:00	Group photo	
17:00-18:00	Move to banquet restaurant	Please follow the guide to the bus
18:00-20:00	Banquet	
Tuesday (October 13)	**Program**	**Notes**
08:00-09:15	**Emerging topics in underwater acoustics**	5 minutes * 15 talks
09:15-09:50	**Poster session & coffee break**	35 minutes
09:50-11:20	**Acoustical Assessment (session)**	15 minutes * 6 talks

11:20-11:40	**Acoustical Assessment (discussion)**	20 minutes
11:40-12:10	**Regional updates & roundtable**	5 minutes updates *4 + 10 minutes discussion
12:10-12:20	Wrap-up	Conference closing
12:20-13:30	Lunch	
14:00-17:00	Site visit	Only for the reserved participations

PROGRAMME

Date: Monday 12 - 9:00 ~ 10:00

OPENING

Dong-Guk Paeng (Chair of conference)

Department of Ocean System Engineering, Jeju National University, Republic of Korea

INVITED TALKS

Chi-Fang Chen

National Taiwan University, Taiwan

Acoustic Density Estimation and Localization of Marine Mammals Using Passive Acoustic Monitoring in Taiwan's Offshore Wind Farm Areas

Songhai Li

Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences, China

Conservation-related Bioacoustic Researches on Cetaceans in Chinese Waters

Date: Monday 12 - 10:00 ~ 11:20

MARINE BIOACOUSTICS

Chair

Songhai Li

Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences, China

Hansoo Kim

Korea Institute of Ocean Science & Technology, Republic of Korea

Large-scale monitoring of coastal odontocetes using multiple static acoustic sensors

Songhai Li
Chinese Academy of Sciences, China

Acoustic Characterization of Representative Sound Producing Fishes in the Northwest Pacific Ocean

Hansoo Kim
Korea Institute of Ocean Science & Technology, Republic of Korea

Unsupervised detection of marine mammal acoustic activity in changing East Siberian Sea soundscape

Siddharth Khopkar
Korea Polar Research Institute, Republic of Korea

Occurrence of Cetaceans at a Proposed Floating Offshore Wind Farm Site Using Passive Acoustic Monitoring

Sujung Kim
Pukyong National University, Republic of Korea

Date: Monday 12 - 13:00 ~ 14:50

SONAR AND SIGNAL PROCESSING

Chair

Gi-hoon Byun

Korea Maritime & Ocean University, Ocean Science and Technology School, Republic of Korea

Ge Yu

Chinese Academy of Sciences, Harbin Engineering University, China

Machine Learning-Based Source Ranging Using Beamformed Data from a Bottom-Mounted Horizontal Line

Gi-hoon Byun
Ocean Science and Technology School, Republic of Korea

Energy Distribution Characteristics of Typical Underwater Noise Fields in the Circular Harmonic Domain for Array Signal Processing

Ge Yu
Chinese Academy of Sciences, Harbin Engineering University, China

Acoustic Monitoring of Antarctic Silverfish Distribution in a Warming Ross Sea: A Random Forest Framework for Ecosystem Sensitivity Assessment

Sara Lee
Korea Polar Research Institute, Republic of Korea

An Acoustic Buoy Self-Noise Suppression Method Based on Spectral Subtraction

Chunlong Huang
Harbin Engineering University, China

Target strength characteristics of live Japanese flying squid

Naizheng Yan
Hokkaido University, Japan

An Analytical Study on Radiated Noise Characteristics of Controllable Pitch Propellers

Jeongwoo Lee
Hanyang University ERICA, Republic of Korea

Date: Monday 12 - 15:20 ~ 16:55

NOISE AND EFFECT

Chair

Tomonari Akamatsu

Research Organization for Nano & Life Innovation, Waseda University, Japan

Changsoo Kim
Department of Ocean System Engineering, Jeju National University, Republic of Korea

High-frequency passive acoustic monitoring of small odontocetes for the environmental assessment of offshore wind farm

Tomonari Akamatsu
Waseda University, Japan

Quantifying Coastal Dolphin Exposure to Noise from Small Vessels

Changsoo Kim
Jeju National University, Republic of Korea

Study on underwater noise monitoring during the construction of suction bucket jacket foundations

Meng-Fan Tsai
National Taiwan University, Taiwan

Acoustic Characteristics of Finless Porpoises (Neophocaena asiaeorientalis) and Their Relationships with Environmental Variables Revealed by Passive Acoustic Monitoring in Korean Coastal Aquaculture Waters

Dongha Kang
National Institute of Fisheries Science, Republic of Korea

Finless Porpoise Occurrence Patterns at the Southwest Offshore Windfarm Using Passive Acoustic Monitoring

Young Geul Yoon
Korea Institute of Ocean Science & Technology, Republic of Korea

Date: Tuesday 13 - 08:00 ~ 09:50

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Chair

Changsoo Kim
Department of Ocean System Engineering, Jeju National University, Republic of Korea

Passive Underwater Target Velocimetry Using Acoustic Vector Sensors and Virtual Interference Array Processing
Bo Wang
Harbin Engineering University, China

Statistical Characteristics of Acoustic Fields in Shallow Water with a Rough Seabed: Theoretical Modeling and Experimental Analysis
Lei Wang
Harbin Engineering University, China

Inversion of Geoacoustic Parameters Using Vertical Coherence of Ocean Ambient Noise
Xiao Rui
Harbin Engineering University, China

Spatial Radiation Pattern and Acoustic Characterization of High-Energy Laser-Induced Underwater Transient Sources
Zhengyang Zhang
Harbin Engineering University, China

Twin Support Vector Machine-Based Discrimination of UUV Interference Tonals and Target-Radiated Tonals
Zheng Shen
Harbin Engineering University, China

Near-Field Interference Suppression Beamforming for Vector Hydrophone Arrays Based on Covariance Matrix Reconstruction
Weiqiang Xia
Harbin Engineering University, China

A Sparse DOA Estimation Algorithm Resistant to Strong Near-Field and Far-Field Interference Based on the Generalized Laplace Prior
Jun Li
Harbin Engineering University, China

Effects of Manual Data Quality Assessment on Ship Classification Performance Using Public Ship Noise Datasets

Hongsun Jeong
Sejong University, Republic of Korea

A Low-Frequency Tonal Direction-of-Arrival Estimation Method for Small-Aperture UUV Sonar Arrays

Xuyan Liu
Harbin Engineering University, China

Background-Normalized Multi-Frame Circular Convolution for Multi-Target Bearing Tracking of Underwater Noise Sources

Yangyang Xue
Harbin Engineering University, China

Delay-Compensated Source Range Estimation of Ship-Radiated Noise Using a Deformed Vertical Array

Minkyu Kim
Korea Maritime & Ocean University, Republic of Korea

Assessment of Interannual Variability in Fishery Resources within an Offshore Wind Farm Using Hydroacoustic Surveys

Taegyeom Hwang
Division of marine production system management, Republic of Korea

DOA Estimation Using MTL-SINN for Passive Sonar Systems

Younghoon Jeong
Seoul National University, Republic of Korea

Shallow-Water Measurement of Underwater Radiated Noise from a Small-Sized USV for Developing an Accredited Underwater Noise Assessment System

Wan-Gu Kim
Korea Institute of Ocean Science & Technology, Republic of Korea

Dechannelized Representation for Passive Sonar Target Classification

Junho Bae
Seoul National University, Republic of Korea

Date: Tuesday 13 - 09:50 ~ 11:40

ACOUSTICAL ASSESSMENT

Chair

Hyoung Sul La

Division of Ocean & Atmosphere Sciences, Korea Polar Research Institute, Republic of Korea

Myounghee Kang
Department of Maritime Police and Production System, Gyeongsang National University, Republic of Korea

Passive Acoustic Monitoring of Polar Soundscapes at KOPRI
Hyoung Sul La
Korea Polar Research Institute, Republic of Korea

Integrated Active and Passive Acoustic Monitoring of Finless Porpoises and Fish Aggregations at a Set-net Fishing Ground in Goseong, South Korea
Myounghee Kang
Gyeongsang National University, Republic of Korea

Temporal and Spatial Variability in Underwater Soundscapes Around the Japanese Coast
Tetsuro Yabuuchi
Asia Air Survey Co.,Ltd, Japan

Distribution and density of Fish Resource in Marine Aggregate Extraction Areas Using Acoustic Surveys and Environmental DNA Analysis
Geunchang park
Pukyong National University, Republic of Korea

Frequency-Dependent Characteristics of Underwater Ambient Noise at Different hydrophone Depths in the Deep East Sea of Korea
Seung-Uk Im
Korea Institute of Ocean Science & Technology, Republic of Korea

Environmental Controls on Low-Frequency Underwater Sound Levels under Changing Sea-Ice Conditions in the East Siberian Arctic Shelf
Moon Ju Jo
Hanyang University ERICA, Republic of Korea

Date: Tuesday 11:40 ~ 12:10

REGIONAL UPDATES AND ROUNDTABLE DISCUSSION

Chair

Dong-Guk Paeng

Department of Ocean System Engineering, Jeju National University, Republic of Korea

Regional representative

Changsoo Kim

Department of Ocean System Engineering, Jeju National University, Republic of Korea

Chi-Fang Chen

National Taiwan University, Taiwan

Tomonari Akamatsu

Research Organization for Nano & Life Innovation, Waseda University, Japan

Songhai Li

Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences, China

ABSTRACTS

INVITED TALKS Monday 12 - 9:10 ~ 10:00

Acoustic Density Estimation and Localization of Marine Mammals Using Passive Acoustic Monitoring in Taiwan's Offshore Wind Farm Areas

Chi-Fang Chen

National Taiwan University, Taiwan

Shashidhar Siddagangaiah

National Taiwan Ocean University, Taiwan

Wei- Chun Hu

National Taiwan University, Taiwan

Jie- Wen Guo

National Taiwan University, Taiwan

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Abstract: Taiwan has set an ambitious target of achieving carbon neutrality by 2050, with offshore wind power anticipated to generate 40–55 GW. The rapid development of offshore wind farms along the Eastern Taiwan Strait (ETS) introduces significant underwater noise from construction activities such as pile driving, increased vessel traffic during surveys and construction, and continuous operational noise over a projected lifespan of 20–30 years. This escalating anthropogenic noise imposes considerable stress on the marine soundscape and its ecological constituents.

Our research focuses on long-term acoustic monitoring at wind farm sites (2014-2025), investigating the changes in sound levels and their ecological consequences, particularly the acoustic behaviors of fish and the critically endangered Taiwanese humpback dolphin (Sousa chinensis taiwanensis). For the first time, our findings reveal clear impacts of pile-driving and operational noise on fish vocalization patterns and demonstrate how vessel traffic and windfarm-related noise influence the acoustic behavior of Taiwanese humpback dolphins. Additionally, we

have developed automated detection algorithms to effectively identify vocalizations of both fish and dolphins from extensive acoustic datasets.

The acoustic datasets were used to estimate marine mammal density and localize individual animals. The results demonstrate the effectiveness of passive acoustic monitoring (PAM) for detecting, localizing, and monitoring marine mammals, providing valuable information on their spatial and temporal distribution.

This research is funded by Taiwan National Science and Technology Council, Unitech Inc., Ørsted Taiwan, Copenhagen Infrastructure Partners (CIP), Taiwan Power Company.

INVITED TALKS Monday 12 - 9:10 ~ 10:00

Conservation-related Bioacoustic Researches on Cetaceans in Chinese Waters

Songhai Li

Marine Mammal and Marine Bioacoustics Laboratory, Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences

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Keywords: Sound production, hearing, passive acoustic monitoring, cetaceans

Abstract: Conservation on cetaceans are highly concerned worldwide in recent decades. Cetaceans are known to produce various sounds frequently, and mainly rely on sounds for communication, emotion expression, mating, foraging, and even navigation. Underwater acoustic environment is one of essential factors to consider in terms of conservation on cetaceans. Therefore, conservation-related bioacoustics researches including but not limited to sound production, hearing, noise effect, and passive acoustic monitoring on cetaceans have been developed over 60

years since 1950s in the world. There are almost 40 cetaceans documented in Chinese waters, including the Yangtze River dolphin and Yangtze finless porpoise unique to the middle and lower reaches of the Yangtze River. The conservation-related bioacoustics researches on cetaceans in Chinese waters were initiated in 1980s with the Yangtze River dolphin and Yangtze finless

porpoise, and being extended to Indo-Pacific humpback dolphins, Bryde's whales, and other cetacean species in the South China Sea, East China Sea, and Yellow Sea. These efforts have shed light on behaviour, ecology, population status, distribution pattern, and so on, for the investigated animals, and benefited protection policy designation and management for the concerned species. This presentation will summarize the achievements of these efforts in the past decades.

MARINE BIOACOUSTICS Monday 12 - 10:00 ~ 11:20

Unsupervised detection of marine mammal acoustic activity in changing East Siberian Sea soundscape

Siddharth Khopkar

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MARINE BIOACOUSTICS

Monday 12 - 10:00 ~ 11:20

Keywords: Passive acoustic monitoring, Acoustic scenes, Unsupervised learning, Vocal phenology, Soundscape ecology, Machine learning

Abstract: The Arctic Ocean is undergoing rapid environmental transformation, with cascading impacts on marine mammal habitat use and acoustic behavior. Passive acoustic monitoring provides a powerful means to track these dynamics, yet methods that can identify biologically relevant acoustic activity in complex, data-scarce recordings without large annotated training sets are still limited. In this study, we combine acoustic indices with unsupervised clustering to detect and characterize marine mammal acoustic presence within the underwater soundscape of the East Siberian Sea, a remote and acoustically understudied Arctic shelf region shaped by seasonal sea ice and extreme photoperiod. We analyze passive acoustic recordings from two non-consecutive annual deployments (2017–2018 and 2019–2020), together comprising over 170,000 one-minute acoustic clips spanning full seasonal cycles. Fourteen indices summarizing energy distribution, temporal variability, and spectral complexity were extracted from each clip and used to identify recurring acoustic states. A two-stage unsupervised clustering framework, combining k-means++ partitioning with Ward hierarchical agglomeration, is used to isolate biologically enriched acoustic regimes containing tonal and frequency-modulated signals consistent with pinniped and cetacean vocalizations, distinguishing them from geophonic states dominated by ice-associated and wind-driven noise, with two-dimensional UMAP embeddings used to visualize regime separation in acoustic feature space. Clustering further reveals how the prevalence and timing of marine mammal acoustic activity shift across the annual cycle, enabling comparison of biologically active periods between contrasting ice conditions in the two deployment years. Beyond classification, the index-based features provide interpretable links between marine mammal acoustic presence and seasonal environmental context, particularly sea ice cover. Our results demonstrate that unsupervised soundscape analysis offers a scalable, non-species-specific screening method for identifying marine mammal acoustic activity in polar seas, providing a foundation for long-term monitoring of Arctic marine mammal phenology under ongoing climate-driven habitat change.

Acknowledgment: This research was supported by a grant from the Korea Institute of Marine Science & Technology Promotion (KIMST), funded by the Ministry of Oceans and Fisheries (RS-2021-KS211500), and by the Korea Polar Research Institute (PE26210).

MARINE BIOACOUSTICS Monday 12 - 10:00 ~ 11:20

Acoustic Characterization of Representative Sound Producing Fishes in the Northwest Pacific Ocean

Hansoo Kim

Sea Power Reinforcement·Security Research Department, Korea Institute of Ocean Science & Technology (KIOST), Republic of Korea

Young Geul Yoon

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MARINE BIOACOUSTICS

Monday 12 - 10:00 ~ 11:20

Keywords: biological fish sounds, vocalization patterns, Passive acoustic monitoring (PAM)

Abstract: Passive acoustic monitoring (PAM) is widely used for observing marine biological sounds to effectively and non-invasively detect and characterize ocean soundscapes and sound-producing organisms. This study synthesizes recent findings on the acoustic characteristics of representative sound-producing fishes in the Northwest Pacific Ocean, including Brown croaker, Small yellow croaker, Pacific cod, and Bluefin gurnard. We compare vocalization patterns reported in recent literature, with particular emphasis on call and chorus characteristics in Brown and Small yellow croakers, and grunt and chorus characteristics in Pacific cod and Bluefin gurnard. As a representative case study, we also examine Brown croaker vocal activity using power spectral density (PSD) analyses across diel and interannual time scales. In-situ recordings were collected continuously for one month in September of 2025 and 2026 at an aquaculture sea-cage using a self-recording hydrophone. Vocalization patterns were compared between years and among diel periods, including sunset, dusk, and midnight, in relation to water temperature and diel variation. Brown croaker vocal activity occurred daily near sunset, while reduced activity was observed when water temperature was below 22 °C, suggesting a decrease in spawning-related sound production under cooler conditions. These results highlight the efficacy of PAM for identifying spawning-related acoustic activity and potential spawning and fishing grounds. This study provides valuable baseline information for monitoring and managing the spawning and fishing grounds of economically important fish species in the Northwest Pacific Ocean.

Acknowledgment: This research was supported by the “AI-based Identification and Classification of Marine Mammal Acoustic Signals for Establishing Marine Protected Areas (PEA0452)” project funded by the Korea Institute of Ocean Science & Technology (KIOST).

MARINE BIOACOUSTICS Monday 12 - 10:00 ~ 11:20

Occurrence of Cetaceans at a Proposed Floating Offshore Wind Farm Site Using Passive Acoustic Monitoring

Sujung Kim

Division of marine production system management, Pukyong National University, Republic of Korea

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Keywords: offshore wind farm, cetaceans, Passive Acoustics Monitoring, click, whistle

Abstract: As global energy demands increase, offshore wind farm development has the potential to provide a valuable source of renewable energy. However, underwater noise and habitat change associated with offshore wind farm construction and operation may affect marine mammals. In this study, we investigated the cetacean occurrence patterns at station AM-3 within an offshore wind farm area using passive acoustic monitoring. Acoustic data were collected from February to July 2023 using a hydrophone with a sampling rate of 384 kHz. The recordings were analyzed using dB-Wav and PAMGuard. Whistles were detected using the Whistle and Moan Detector and classified using the Real-time Odontocete Call Classification Algorithm (ROCCA) with the Temperate Pacific Classifier (TPC) package. Clicks were detected using the PAMGuard Click Detector. Automatically detected and classified signals were then confirmed manually in dB-Wav by examining the waveforms and spectrograms. The analysis detected acoustic signals from

several cetacean species, including common dolphins, bottlenose dolphins, Risso's dolphins, killer whales, and Pacific white-sided dolphins. Whistle detections peaked in July, and whistles classified as bottlenose dolphins and Risso's dolphins were detected most frequently. Click detections peaked in March and clicks classified as common dolphins were detected most frequently. Common dolphin clicks were detected on 140 days, while Risso's dolphin clicks were detected on 126 days during the monitoring period. These findings provide baseline information on the occurrence and temporal patterns of marine mammals in the offshore wind farm area. Further analyses of diel occurrence patterns are required, and sighting surveys should be combined with long-term passive acoustic monitoring to improve assessments of cetacean distribution and abundance.

MARINE BIOACOUSTICS Monday 12 - 10:00 ~ 11:20

Large-scale monitoring of coastal odontocetes using multiple static acoustic sensors

Songhai Li

Marine Mammal and Marine Bioacoustics Laboratory, Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences

Lijun Dong

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Mingming Liu

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Zirui You

Marine Mammal and Marine Bioacoustics Laboratory, Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences

Abstract: Passive acoustic monitoring (PAM) is increasingly being adopted as a non-invasive method for the assessment of ocean ecological dynamics. PAM is an important sampling approach for acquiring critical information about marine mammals, especially in areas where data are lacking and where evaluations of threats for vulnerable populations are required. The Indo-Pacific

humpback dolphin (IPHD, *Sousa chinensis*) and finless porpoise (IPFP, *Neophocaena phocaenoides*) are coastal odontocetes which inhabit tropical and warm-temperate waters from the eastern Indian Ocean throughout Southeast Asia to central China. The distribution pattern of these animals around the Hainan Island (China) is unclear, but evidences for a resident population of both the species have recently emerged. An array of passive acoustic platforms was deployed at depths of 10–20 m (the preferred habitat of humpback dolphins, and potential habitat of finless porpoises), across sites covering more than 100 km of coastline. In this study, we explored the spatiotemporal patterns of distribution, acoustic behavior, and potential noise effect of these species by using the PAM data through supervised machine learning algorithms. Our work demonstrates that the distribution and habitat use of the coastal and resident odontocetes can be monitored over a large spatiotemporal scale, using an array of passive acoustic platforms and a data analysis protocol that includes both machine learning techniques and spectrogram inspection, and potential noise effect was evaluated.

SONAR AND SIGNAL PROCESSING Monday 12 - 13:00 ~ 14:50

Energy Distribution Characteristics of Typical Underwater Noise Fields in the Circular Harmonic Domain for Array Signal Processing

Fan DU

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Key Laboratory of Marine Information Acquisition and Security, Harbin Engineering University, Ministry of Industry and Information Technology, China
State Key Laboratory of Acoustics and Marine Information, Chinese Academy of Sciences, China

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Yanli CHEN

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Keywords: circular harmonic domain, noise energy distribution, sonar array signal processing, uniform circular array

Abstract: To address the requirements of sound field modeling and noise suppression in the circular harmonic domain, this paper adopts the uniform circular array as the typical array configuration and systematically investigates the modal-domain energy distribution characteristics

of typical underwater ambient noise fields to satisfy the practical application requirements of sound field modeling, interference suppression and array signal processing in the circular harmonic domain. Three classic marine ambient noise field models, includes the spatially white noise field, cylindrically isotropic noise field and spherically isotropic noise field, are selected as the research objects. Relying on the sound field decomposition theory in the circular harmonic domain, this paper theoretically derives the mathematical expressions of energy distribution for the three types of noise fields under circular harmonic modes of different orders, and deeply explores the inherent correlation mechanism among array configuration, modal order, and noise energy distribution. By establishing a simulation model of underwater noise fields and conducting comparative simulation experiments, this study quantitatively analyzes the variation laws of energy concentration, attenuation, and diffusion of the three noise fields across different circular harmonic modal orders. The research outcomes not only enrich and perfect the theoretical system of underwater noise field modeling within the circular harmonic domain, but also provide solid theoretical support reasonable parameter selection criteria and effective algorithm design references for key underwater array signal processing technologies including circular harmonic domain adaptive beamforming, passive target direction-of-arrival estimation, spatial noise suppression and simulation modeling of complex marine sound fields. Analysis of underwater ambient noise characteristics is fundamental to the design and performance optimization of array signal processing algorithms.

Acknowledgment: This research was supported by the State Key Laboratory of Acoustics and Marine Information, Chinese Academy of Sciences (SKLA202507) .

SONAR AND SIGNAL PROCESSING Monday 12 - 13:00 ~ 14:50

Acoustic Monitoring of Antarctic Silverfish Distribution in a Warming Ross Sea: A Random Forest Framework for Ecosystem Sensitivity Assessment

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Keywords: Antarctic silverfish, acoustic monitoring, random forest, Ross Sea, climate change, ecosystem sensitivity

Abstract: Antarctic silverfish (*Pleuragramma antarcticum*) is an ecologically pivotal mesopelagic fish in the Ross Sea, where rapid ocean warming is reshaping the physical and acoustic environment structuring marine ecosystems. Using active hydroacoustic data from five cruises (2018–2023), we developed a random forest regression model to predict the nautical area scattering coefficient (NASC) of Antarctic silverfish from kriged temperature and salinity fields, with fish acoustically identified using a multi-frequency target strength criterion (80–250 mm total length). The final model achieved an out-of-bag R^2 of 0.862, with longitude and salinity as primary predictors. In scenario-based sensitivity experiments referencing IPCC

AR6 SSP pathways, predicted NASC increased under most warming perturbations, with the largest response under SSP5-8.5 (+100%). This acoustic framework supports ecosystem monitoring within the Ross Sea Marine Protected Area under a changing ocean.

Acknowledgement: This work was supported by Korea Institute of Marine Science & Technology Promotion (KIMST) grant funded by the Ministry of Oceans and Fisheries (RS-2022-KS221661).

SONAR AND SIGNAL PROCESSING Monday 12 - 13:00 ~ 14:50

Machine Learning-Based Source Ranging Using Beamformed Data from a Bottom-Mounted Horizontal Line Array

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Keywords: range estimation, horizontal line array, beamforming, machine learning

Abstract: Previous machine-learning approaches to source range estimation have used normalized sample covariance matrices (SCMs) constructed from a vertical line array to learn the statistical relationship between observed acoustic fields and source range, demonstrating promising performance. However, data-driven source ranging using a horizontal line array (HLA) remains comparatively underexplored. In this study, we develop a machine-learning framework for source-range estimation using a bottom-mounted HLA. Our analysis shows that using SCMs constructed from HLA results in substantial degradation in range-estimation performance when the source bearings differ between the training and test datasets. This degradation occurs because SCMs encode not only range-dependent propagation structure but also bearing-dependent phase differences. To address this limitation, we introduce an input structure that uses the dominant beamformer output at each frequency. By compensating for bearing-dependent phase differences before feature extraction, the proposed structure removes bearing dependence while preserving range-dependent broadband interference structure and providing array gain. The results demonstrate robust source-range estimation under mismatched source-bearing conditions. The method is validated through numerical simulations and experimental data.

SONAR AND SIGNAL PROCESSING Monday 12 - 13:00 ~ 14:50

An Acoustic Buoy Self-Noise Suppression Method Based on Spectral Subtraction

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Keywords: self-noise and suppression, acoustic buoy, sea surface waves, noise prediction, deep learning algorithm

Abstract: Ambient noise contains a vast amount of information such as the sound speed of seawater, acoustic parameters of seabed sediment, and the behaviors of marine animals. One of the significant means for ocean ambient noise acquisition is achieved through acoustic buoys. However, a significant challenge exists that the excitation of sea surface waves is transmitted to the underwater hydrophone through the cable, causing the hydrophone to undergo significant vertical undulating motion. Thus, introduces strong motion self - noise, severely affecting the effectiveness of ocean ambient noise acquisition. To address this problem, this study proposes an acoustic buoy self-noise suppression method based on spectral subtraction. We collected the self - noise of acoustic buoys, constructed a dataset, and completed the training of an acoustic buoy self - noise prediction network based on the PatchMixer optimized model. Thus, we achieved the prediction of the acoustic buoy self - noise under different sea surface excitations. Combining sea surface wave data with the acoustic buoy self - noise prediction model, the self - noise suppression is achieved by using the spectral subtraction method.

Acknowledgment: This work was financially supported by the National Natural Science Foundation of China (Grant No.12304498) and project ZR2023QA116 supported by Shandong Provincial Natural Science Foundation.

SONAR AND SIGNAL PROCESSING Monday 12 - 13:00 ~ 14:50

Target strength characteristics of live Japanese flying squid

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Keywords: Japanese flying squid; *Todarodes pacificus*; target strength; broadband acoustics; acoustic scattering; mantle length; acoustic biomass estimation

Abstract: The Japanese flying squid *Todarodes pacificus* is an important fishery resource in the northwestern Pacific. Accurate target strength (TS) estimates are essential for acoustic biomass estimation; however, squid TS is highly variable because of their flexible body structure and changes in swimming orientation. This study examined the TS of live Japanese flying squid using tethered-individual measurements and free-swimming group measurements in seawater tanks. Broadband measurements of tethered-individual measurements showed substantial variation in TS spectra among squid, although relatively high TS values were often observed near 120 kHz. In free-swimming group measurements at 38, 70, 120, and 200 kHz, mean TS increased clearly with mantle length, whereas the maximum difference among frequencies was 3.1 dB. Free-swimming

group squid TS values were generally higher than tethered-individual squid TS by up to approximately 5 dB. The higher values obtained from free-swimming group squid may reflect differences between constrained and freely swimming orientations. These results indicate that body size strongly determines mean TS, while swimming orientation and individual morphology contribute to TS variability. Combining tethered-individual and free-swimming group measurements is therefore useful for improving acoustic biomass estimation of Japanese flying squid.

SONAR AND SIGNAL PROCESSING Monday 12 - 13:00 ~ 14:50

An Analytical Study on Radiated Noise Characteristics of Controllable Pitch Propellers

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Keywords: Controllable pitch propellers(CPP), Radiated noise, speed-dependent, SAVEX15

Abstract: Controllable pitch propellers (CPPs) regulate vessel speed by adjusting blade pitch angles while maintaining a nearly constant engine rotational speed. Despite their widespread use, interpreting the acoustic signatures of CPP vessels remains challenging because conventional acoustic identification monitoring frameworks are heavily tailored to fixed-pitch propeller (FPP) ships, where vessel speed is directly proportional to shaft RPM. Operating a CPP at off-design pitch conditions under constant RPM alters local hydrodynamics, leading to distinct speed-dependent acoustic behaviors and cavitation characteristics. In this study, field acoustic data collected during the Shallow-water Acoustic Variability Experiment 2015 (SAVEX15) are analyzed to examine the underwater radiated noise signatures and cavitation-related acoustic features of a CPP-equipped vessel across various transit speeds. The analysis evaluates speed-dependent variations in propulsion-related spectral components, broadband noise levels, and amplitude modulation behaviors under constant engine speed conditions. By comparing the spectral features obtained at different speeds, this study characterizes the unique acoustic

variations induced by pitch angle adjustments and discusses their physical implications on cavitation modulation. These observational findings provide practical insights for passive acoustic detection and identification of CPP-equipped vessels.

NOISE AND EFFECTS Monday 12 - 15:20 ~ 16:35

High-frequency passive acoustic monitoring of small odontocetes for the environmental assessment of offshore wind farm

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Abstract: Offshore wind energy is one of the major sources of renewable energy, and Asian coastal and offshore waters have substantial potential for wind energy development because of their favorable wind conditions. At the same time, these waters support high marine biodiversity and provide important habitats for many protected species. Small odontocetes are upper-level predators whose occurrence reflects the availability of prey and broader ecosystem conditions, making them potentially useful indicators of marine ecosystems. They frequently produce high-frequency ultrasonic pulses, known as echolocation clicks, which can be automatically detected using dedicated acoustic instruments. Here, we present examples of high-frequency passive acoustic monitoring using an automatic click detector (A-tag) for assessing the effects of offshore wind turbine construction and for characterizing baseline habitat use by small odontocetes at a floating offshore wind farm site. During wind turbine construction, the acoustic presence of finless porpoises decreased markedly at the impact site, whereas no corresponding decrease was detected at a control site 2 km away. A Before-After-Control-Impact (BACI) design was used to distinguish construction-related effects from natural temporal variation. One year after construction, acoustic occurrence at the turbine site was comparable to that at the control site, with similar temporal patterns. Baseline monitoring off Fukushima, Japan, also revealed frequent occurrence of phocoenids in coastal waters and sporadic occurrence of delphinids in offshore waters, consistent with previous knowledge of their habitat use. These results demonstrate that high-frequency passive acoustic monitoring provides an effective means of characterizing baseline habitat use by small odontocetes and assessing their responses to anthropogenic activities. Appropriate survey designs, such as BACI, are essential for distinguishing anthropogenic impacts from natural spatial and temporal variability in habitat use.

NOISE AND EFFECTS Monday 12 - 15:20 ~ 16:55

Quantifying Coastal Dolphin Exposure to Noise from Small Vessels

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Keywords: noise and effect, marine bioacoustics, marine reusable energy, sonar and signal processing, management and policy

Abstract: The global expansion of coastal dolphin tourism utilizing small vessels has raised concerns regarding the acoustic impact of engine noise on marine mammals. To evaluate this impact accurately, quantitative monitoring correlates underwater noise levels with the precise spatial proximity of dolphins and vessels is essential. However, a significant methodological challenge exists that small tourism vessels frequently operate without Automatic Identification System (AIS) transponders, making it difficult to acquire accurate, real-time positioning data in the field. To address this gap, this study proposes a novel monitoring framework utilizing unmanned aerial vehicles (UAVs) and passive acoustic monitoring. We conducted aerial surveys to film dolphin pods and tourism vessels simultaneously. By integrating the pixel data from the aerial imagery with the drone's onboard GPS logs, we georeferenced the footage to derive the absolute longitude and latitude coordinates of both the dolphins and the vessels. Synchronously, underwater hydrophones were deployed to measure real-time vessel noise levels.

NOISE AND EFFECTS

Monday 12 - 15:20 ~ 16:55

Acknowledgment: This research was supported by Basic Science Research Program to the Research Institute for Basic Sciences (RIBS) of Jeju National University through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2019R1A6A1A10072987).

NOISE AND EFFECTS Monday 12 - 15:20 ~ 16:55

Study on underwater noise monitoring during the construction of suction bucket jacket foundations

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Keywords: offshore wind power, suction bucket jacket, underwater noise

Abstract: Taiwan is rich in wind energy resources. providing excellent conditions for offshore windfarm development. Consequently, global offshore wind developer Ørsted selected this region to advance the Greater Changhua Offshore Wind Project with a total installed capacity of 1.82 GW. Phase 2 of this project is the first commercial-scale offshore wind project to adopt Suction Bucket Jacket (SBJ) underwater foundations in Taiwan. By eliminating traditional piling procedures, it effectively reduces seabed disturbance, underwater noise, and potential impacts on the marine ecological environment.

Monitoring results show that the equivalent continuous sound level (L_{eq}) measured during installation ranged approximately between 126 and 138 dB, representing an increase of about 5 dB compared to background noise before installation. Further cross-analysis between offshore installation procedures and underwater noise monitoring results revealed that the median sound pressure levels across different installation steps were all between 133 and 135 dB. The variations

among procedures were insignificant, indicating that underwater noise generated across different installation stages was similar, with no specific procedure causing noticeably higher noise.
Overall, By eliminating traditional piling procedures for jacket foundations, underwater noise exposure during construction can be significantly reduced. Therefore, its potential impact on marine organisms and the marine environment is expected to be lower than that of conventional pile-driving methods.

Acknowledgements: This study is supported by Ørsted Taiwan Ltd.

NOISE AND EFFECTS Monday 12 - 15:20 ~ 16:55

Acoustic Characteristics of Finless Porpoises (*Neophocaena asiaeorientalis*) and Their Relationships with Environmental Variables Revealed by Passive Acoustic Monitoring in Korean Coastal Aquaculture Waters

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NOISE AND EFFECTS

Monday 12 - 15:20 ~ 16:55

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Keywords: finless porpoise, passive acoustic monitoring, Generalized Additive Model, echolocation clicks, aquaculture

Abstract: The finless porpoise (*Neophocaena asiaeorientalis*) is frequently observed in Korean coastal waters but remains vulnerable to bycatch and habitat disturbance. This study used passive acoustic monitoring to examine its acoustic characteristics and activity patterns near cage aquaculture farms from September to October 2021. In total, 372,707 clicks and 175,119 click trains were detected. Mean acoustic parameters included a peak frequency of 122.0 ± 11.1 kHz, 3 dB bandwidth of 15.4 ± 12.0 kHz, 10 dB bandwidth of 45.3 ± 16.1 kHz, and ICI of 39.0 ± 34.8 ms. Click activity showed a clear diel pattern, with higher activity during the early morning and nighttime. Generalized additive model analysis indicated significant non-linear relationships between click activity and tide, temperature, salinity, and hour. Click activity decreased at higher temperatures and lower salinity, whereas tide had a limited effect. These findings provide baseline data for acoustic deterrent development and bycatch mitigation in Korean coastal waters.

Acknowledgment: This research was supported by KIOST (grant no. EA0422), NRF/MSIT (grant no. RS-2025-00520657), and NIFS (grant no. R2026008).

NOISE AND EFFECTS Monday 12 - 15:20 ~ 16:55

Finless Porpoise Occurrence Patterns at the Southwest Offshore Windfarm Using Passive Acoustic Monitoring

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NOISE AND EFFECTS

Monday 12 - 15:20 ~ 16:55

Keywords: offshore wind farm, finless porpoise, echolocation click, passive acoustic monitoring

Abstract: As offshore wind farms (OWFs) continue to expand globally, there is a growing need for continuous monitoring to understand the impacts of underwater noise associated with these operations on marine life. In South Korea, the recent expansion of OWFs has raised the need to establish baseline information on the occurrence and habitat use of finless porpoises in these waters. To address this, the present study examined the temporal occurrence patterns of finless porpoises by conducting passive acoustic monitoring (PAM) for approximately 10 days at the Southwest Offshore Windfarm in South Korea. The acoustic data showed that finless porpoise detections were more frequent during high tide periods, indicating that tidal conditions may play an important role in shaping their occurrence within the study area. These findings provide baseline information on the temporal occurrence of finless porpoises in an operating OWF and contribute to future assessments of the ecological effects of offshore wind farm development on marine mammals.

Acknowledgment: This work was supported by "National Marine Ecosystem Monitoring Program" funded by the Ministry of Oceans and Fisheries (MOF) of the Republic of Korea.

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Passive Underwater Target Velocimetry Using Acoustic Vector Sensors and Virtual Interference Array Processing

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Keywords: acoustic vector sensor (AVS), passive velocity estimation, virtual interference array, broadband multipath interference, high-resolution spectral estimation.

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Abstract: Passive velocity estimation of underwater moving targets is essential for underwater surveillance, target tracking, and ocean acoustic sensing. Conventional passive velocimetry methods generally rely on multi-element arrays and beamforming techniques, resulting in increased system complexity, deployment cost, and operational constraints. Developing compact and robust velocity estimation techniques using limited sensing resources remains a challenging problem. To address this issue, a passive velocity estimation method based on acoustic vector sensor (AVS) and virtual interference array processing is proposed. The method exploits broadband multipath interference structures contained in both sound-pressure and particle-vibration-velocity measurements. Gaussian-weighted coherent broadband fusion is first employed to enhance interference coherence across frequencies. Temporal-delay interference processing is then introduced to construct a vector virtual interference array in the delay domain. By establishing the relationship between target radial velocity and interference phase evolution, velocity estimation is reformulated as a subspace spectral estimation problem. A vector MUSIC-like velocity spectrum and a Viterbi-based temporal tracking strategy are subsequently developed to improve estimation resolution and continuity. Numerical simulations based on the KRAKEN acoustic propagation model demonstrate that the proposed method can achieve stable and high-resolution velocity estimation under broadband multipath propagation conditions. Compared with conventional pressure-only interference processing, the incorporation of vector acoustic information provides improved estimation accuracy, robustness, and interference suppression capability. The proposed framework combines the advantages of vector acoustic sensing and virtual array processing, enabling high-resolution passive velocimetry using compact sensing systems. The method provides a promising approach for underwater target motion analysis and may facilitate future developments in AVS-based passive localization and tracking.

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Statistical Characteristics of Acoustic Fields in Shallow Water with a Rough Seabed: Theoretical Modeling and Experimental Analysis

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Keywords: seabed scattering, acoustic fluctuation, normal mode theory, envelope distribution, scintillation index, spatiotemporal correlation

Abstract: To address the problem of acoustic field fluctuations induced by a rough seabed, this paper systematically investigates acoustic field statistical characteristics in a sand-wave area of the South China Sea through a combination of theoretical modeling and experimental analysis. In the modeling part, a normal-mode-based method is proposed that avoids Monte Carlo simulations and accounts for omnidirectional single scattering. Numerical simulations show that the influence of the random acoustic field increases with propagation distance, and its spatial correlation coefficient exhibits oscillatory decay in both horizontal longitudinal and vertical directions. The model predictions agree well with measured vertical correlation data of the ocean acoustic field. In the experimental part, stationary acoustic fluctuation data from the same area are analyzed. Results show that as frequency increases, the Rice envelope distribution of the total sound pressure evolves from Gaussian to Rayleigh, and the scintillation index increases. This indicates that the random acoustic field takes the dominant position, thereby affecting the spatiotemporal characteristics of the total sound field. The two studies mutually validate each other, revealing the influence of rough

seabed scattering on shallow-water acoustic field statistics and providing guidance for hydrophone array design and signal processing.

Acknowledgment: This research was supported by Open Fund Project of National Key Laboratory of Underwater Acoustic Technology (Grant No. 2023-JCJQ-LB072-02).

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Inversion of Geoacoustic Parameters Using Vertical Coherence of Ocean Ambient Noise

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EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

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Keywords: Ocean Ambient Noise, Vertical Coherence Coefficient, Sensitivity Analysis, V Geoacoustic Parameter Inversion

Abstract: In the modeling of realistic ocean ambient noise fields, the reciprocity principle is often employed to reduce computational complexity. However, when the sound speed profile and seabed topography are non-uniform, the particle velocity field cannot be directly handled by conventional reciprocity-based methods due to its directionality. In this paper, based on ray acoustics and the reversibility of ray paths, the arrival angle of each ray is corrected so that the particle velocity field can also be efficiently computed using the reciprocity principle. On this basis, a vector field model of ocean ambient noise is established. Through numerical simulations, the influence of the noise source integration radius on the depth-dependent profiles of sound pressure and vertical particle velocity is analyzed, and the sensitivity of the vertical spatial coherence function of the vertical particle velocity to geoacoustic parameters is examined. Furthermore, by incorporating measured vertical coherence data of ocean ambient noise, the seabed geoacoustic parameters are retrieved, and the reliability of the inversion results is validated using acoustic propagation data.

Acknowledgment: This research was supported by the National Natural Science Foundation of China (Grant No. 12204128).

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Spatial Radiation Pattern and Acoustic Characterization of High-Energy Laser-Induced Underwater Transient Sources

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EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Keywords: laser-induced breakdown, underwater sound source, transient acoustic pulses, sonar and signal processing, high-energy pulsed laser

Abstract: Laser-induced acoustic pulses feature non-contact excitation, high intensity, and ultra-wideband characteristics, showing immense potential for high-resolution underwater sensing. Yet, their spatial distribution and potential contribution to the marine soundscape require quantitative evaluation. We induced transient underwater acoustic signals using a high-energy pulsed laser and conducted high-resolution multi-angle spatial mapping utilizing a high-sensitivity hydrophone. The experiment analyzed the time-domain waveforms, spectral features, and spatial energy distribution within the 120 kHz bandwidth. Results demonstrate the source generates transient pulses with an extremely high sound pressure level near the breakdown point. Influenced by laser beam propagation and plasma morphology, the sound field exhibits a distinct, asymmetrical directional distribution. Although the pulses are inherently ultra-wideband, energy within the 120 kHz range is concentrated and highly deterministic in spatial angle. This research bridges the gap in quantitatively describing the spatial radiation pattern of high-energy laser-induced sound sources. Featuring extremely short signal durations and controllable spatial distribution, this source provides high-resolution detection while yielding an environmental noise footprint potentially easier to mitigate than continuous sources. This study provides crucial experimental evidence for the future green application of laser acoustic technology in environmentally sensitive marine areas.

Acknowledgment: This research was supported by Hainan Province Science and technology Special Fund, China (Grant No. ZDYF2024SHFZ086).

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Twin Support Vector Machine-Based Discrimination of UUV Interference Tonals and Target-Radiated Tonals

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EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

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Keywords: unmanned underwater vehicle, platform interference tonals, target-radiated tonals, twin support vector machine

Abstract: Low-frequency tonal components provide important information for passive underwater detection, but determining their origins remains difficult on an unmanned underwater vehicle (UUV). During UUV operation, unknown interference tonals generated by platform self-noise may coexist with target-radiated tonals in the same low-frequency band. Because both types of tonals appear as concentrated narrowband peaks, they cannot be reliably distinguished based only on frequency location. This paper proposes a multi-domain feature method for discriminating these two types of tonals. Sea-trial data are first analyzed to examine their differences in temporal variation, array-element response, and frequency variation, and an array-received signal model is subsequently established. After tonal components within the observation band are uniformly detected, four feature groups are constructed, including near-/far-field matching, spatial energy distribution, time-frequency state fluctuation, and frequency-domain motion response. These features describe tonal differences from temporal, spatial, and spectral perspectives while retaining the detected components. A twin support vector machine is then used to learn the discrimination boundary between the two tonal classes, providing an effective solution for small-sample tonal classification. Simulation and sea-trial experiments show that the proposed features produce clear separability between platform-interference and target-radiated tonals under varying motion states, propagation conditions, and noise levels. Comparative results further demonstrate the classification performance and practical applicability of the proposed method across different datasets and operating conditions. Unlike approaches that suppress near-field components before detection, the proposed method performs signal-level analysis after tonal extraction without modifying the received signals. The resulting tonal labels enable more targeted use of the received signals in subsequent processing stages.

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Acknowledgment: This research was supported by the National Natural Science Foundation of China (Grant No. 62571153).

EMERGING TOPICS IN UNDERWATER ACOUSTICS

Tuesday 13 - 08:00 ~ 09:50

Near-Field Interference Suppression Beamforming for Vector Hydrophone Arrays Based on Covariance Matrix Reconstruction

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Keywords: adaptive beamforming, covariance matrix reconstruction, near-field interference suppression, interference steering vector correction, vector hydrophone array

Abstract: To address the degradation of far-field target detection performance caused by strong near-field interference in ocean environments, this paper introduces a vector hydrophone array and proposes a covariance-matrix-reconstruction-based beamforming method for near-field interference suppression. Unlike conventional pressure sensor arrays, the vector hydrophone array can jointly exploit acoustic pressure and orthogonal particle velocity components, thereby providing additional spatial information for interference characterization and suppression. Based on the acoustic pressure–particle velocity relationship under spherical-wave propagation, the vector-array steering vector of near-field interference is constructed, and a joint acoustic field model containing both far-field plane-wave interferences and near-field spherical-wave interferences is established. To reduce the influence of steering vector mismatch on the accuracy of interference covariance matrix reconstruction, a subspace projection method is employed to correct the interference steering vectors. The corrected steering vectors are then combined with the sample covariance matrix to estimate the interference powers. Meanwhile, the noise power is estimated using a prewhitening filtering method. On this basis, an interference-plus-noise covariance matrix containing both far-field and near-field interference components is reconstructed. Furthermore, the reconstructed covariance matrix is used to formulate a convex optimization problem for correcting the steering vector of the signal of interest, and the robust adaptive beamforming weight vector is subsequently obtained. Simulation results demonstrate that the proposed method can effectively suppress strong near-field interference while maintaining a distortionless response toward the far-field target. The interference suppression depth exceeds 35 dB. In addition, under array element gain-phase errors, random steering vector mismatch, the proposed method still maintains good robustness and stable output performance.

Acknowledgment: This research was supported by the National Natural Science Foundation of China (Grant No. 62301183).

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A Sparse DOA Estimation Algorithm Resistant to Strong Near-Field and Far-Field Interference Based on the Generalized Laplace Prior

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Tuesday 13 - 08:00 ~ 09:50

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Keywords: direction of arrival, near-field interference suppression, generalized Laplace prior, spiky and heavy-tailed distribution

Abstract: Near-field (NF) interference generated by the maneuvering of small-scale underwater platforms degrades the performance of direction-of-arrival (DOA) estimation algorithms for far-field (FF) targets. Moreover, when FF strong targets and FF weak targets are closely spaced in spatial angle, and their signal-to-noise ratios (SNRs) differ significantly, the strong target signals can easily mask the weak ones, manifesting as strong far-field interference, posing a challenge to

the recovery of the weak target signals. To address these issues, this paper proposes a sparse Bayesian learning algorithm based on the generalized Laplace prior to effectively resolve closely spaced FF strong and weak targets in the presence of strong NF interference. First, by introducing an NF spherical wave propagation model into the overcomplete dictionary, the proposed algorithm effectively suppresses the adverse impact of strong NF interference on FF DOA estimation. To further enhance the DOA estimation performance of the algorithm, the generalized Laplace prior is constructed. This approach not only enhances the sparsity constraint on the FF target signals, effectively improving the DOA estimation accuracy under low SNR conditions, but its heavy-tailed structure also significantly enhances the robustness of the algorithm in the presence of strong interference. Numerical simulations and experimental results demonstrate the effectiveness of the proposed method. Compared to other baseline DOA estimation methods, the proposed method exhibits better DOA estimation performance in the presence of strong NF and FF interference.

Acknowledgment: This research was supported by the National Natural Science Foundation of China (Grant No. 62301183).

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Effects of Manual Data Quality Assessment on Ship Classification Performance Using Public Ship Noise Datasets

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Keywords: Passive sonar, Time–frequency analysis, Data quality evaluation, Ship classification

Abstract: Most machine learning approaches for ship classification with publicly available acoustic datasets, focus on improving classification performance using the broad-category labels originally assigned by the dataset providers. However, in real underwater environments, ship-radiated noise recorded by a hydrophone is affected by multipath propagation and is masked by ambient noise. Meaning that data quality may fluctuate in both the time and frequency domains, even for a single recording. To address this issue, this study conducts a detailed manual examination of the DeepShip and ShipsEar datasets using LOFAR, DEMON, spectral, and spectrogram-based analysis methods to more accurately evaluate data reliability. Based on the results of this data quality assessment, a more reliable dataset is constructed by applying refined labels. The results of the feature analysis demonstrate that the dataset with refined labeling provides improved classification performance compared with the dataset based on the original labeling.

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Acknowledgment: This work was supported by the Korea Research Institute for Defense Technology Planning and Advancement under Grant KRIT-CT-22-052 (Physics-guided Intelligent Sonar Signal Detection Research Laboratory, 2026), funded by the Korean Government Defense Acquisition Program Administration.

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Tuesday 13 - 08:00 ~ 09:50

A Low-Frequency Tonal Direction-of-Arrival Estimation Method for Small-Aperture UUV Sonar Arrays

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Keywords: small-aperture array, low-frequency tonal, direction-of-arrival estimation, sparse Bayesian learning

Abstract: Unmanned Underwater Vehicles (UUVs) are limited in size, restricting the aperture of their onboard sonar arrays. In passive sonar detection, low-frequency tonals radiated by underwater targets are of great interest for long-range detection due to their low propagation loss. However, the wavelengths of these tonals are often comparable to, or even larger than, the array aperture on a UUV. This physical mismatch leads to insufficient spatial sampling, severely degrading the bearing resolution of traditional Direction-of-Arrival (DOA) estimation algorithms and making high-resolution bearing estimation of distant weak targets extremely challenging. To address this problem, we exploit the turning motion of the UUV to obtain additional spatial gain by accumulating observations at multiple headings. The cross-spectra between all element pairs under different headings are constructed as an extended observation vector. A Sparse Bayesian Learning (SBL) framework is then employed, wherein an over-complete dictionary matched to the turning-array geometry is established. This formulation automatically promotes sparsity and does not require prior knowledge of the source number. Simulations and experimental results demonstrate that the proposed method achieves high-resolution bearing estimation of low-frequency tonals under small-aperture array conditions, significantly outperforming conventional beamforming (CBF), Minimum Variance Distortionless Response (MVDR), Multiple Signal Classification (MUSIC), and the standard SBL approach without turning-array processing. The method provides a practical solution for low-frequency tonal bearing estimation on size-limited UUV platforms.

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Acknowledgment: This research was supported by the National Natural Science Foundation of China (Grant No. 62571153).

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Background-Normalized Multi-Frame Circular Convolution for Multi-Target Bearing Tracking of Underwater Noise Sources

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Keywords: Underwater noise source tracking, Circular convolution, Particle filter, Vector hydrophone.

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Abstract: Passive acoustic monitoring of underwater noise requires robust tracking of moving acoustic sources under fluctuating and non-uniform backgrounds. For a single vector hydrophone, multi-target bearing-only tracking commonly uses acoustic-intensity-based bearing probability distributions, but these measurements may contain discrete peaks, random fluctuations, and uneven background responses, leading to unstable particle likelihood evaluation. This paper proposes a background-normalized multi-frame circular-convolution track-before-detect method for underwater noise source tracking. The method directly processes bearing probability distributions without frame-wise peak extraction. A Gaussian kernel is constructed in the circular bearing domain and convolved with each frame to suppress isolated fluctuations, enhance local bearing responses, and preserve angular periodicity. The convolution responses along particle-predicted bearing trajectories are accumulated over multiple frames to form a robust likelihood measure. A local background normalization strategy further estimates the mean and variance of convolution responses within a local temporal window, so likelihood evaluation depends on relative response saliency rather than absolute amplitude. Simulations and experimental data show improved robustness and tracking continuity in underwater noise environments.

Acknowledgment: This research was supported by the National Natural Science Foundation of China (Grant No. 12204128).

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Tuesday 13 - 08:00 ~ 09:50

Assessment of Interannual Variability in Fishery Resources within an Offshore Wind Farm Using Hydroacoustic Surveys

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Keywords: Hydro-acoustic survey, Offshore wind farm, dB-difference, Environmental DNA (eDNA)

Abstract: The construction and operation of offshore wind farms may affect marine habitats and the distribution of marine organisms, highlighting the need for long-term and quantitative monitoring. This study evaluated fish distribution in offshore wind farm areas through seasonal hydroacoustic surveys conducted from 2021 to 2025 at wind farm and reference sites in the southwestern waters of Korea and the Hallym coastal waters off Jeju Island. Acoustic data were collected using scientific echosounders operating at 38 and 120 kHz and analyzed using Echoview version 12. Fish echoes were discriminated using the dB-difference method. Environmental DNA

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(eDNA) analysis was additionally conducted to characterize species composition and diversity. In the southwestern waters, fish detections were limited in 2021 and 2022, whereas small fish schools were observed mainly near the surface in 2023. In 2024, fish were distributed primarily near the bottom in spring but throughout the water column in summer. In the Hallym waters off Jeju Island, fish schools were observed near the bottom at the wind farm site and throughout the water column at the reference site in 2021. In 2022 and 2023, fish detections were limited to specific depth ranges and survey transects. Horizontal fish distribution also showed no consistent pattern, as differences in NASC between the wind farm and reference sites varied by year and season. Overall, fish distribution appeared to be influenced by a combination of seasonal and region-specific environmental conditions rather than by the presence of offshore wind farms alone. These findings highlight the need for long-term, repeated monitoring that includes both wind farm and reference sites and integrates hydroacoustic surveys with eDNA analysis to accurately assess the ecological effects of offshore wind farm development.

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DOA Estimation Using MTL-SINN for Passive Sonar Systems

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Keywords: underwater acoustic DOA estimation; System-Informed Neural Network; passive sonar; Deep learning; Sparse reconstruction

Abstract: Direction-of-arrival (DOA) estimation is a key task in passive sonar system. Classical approaches, including CBF, MVDR, and MUSIC, as well as sparse methods such as CS and SBL, have been widely studied. However, these methods involve inherent trade-offs among angular resolution, robustness, computational complexity, and reliance on prior assumptions. Deep-learning-based methods provide fast inference and high angular resolution but often show limited generalization. This paper proposes MTL-SINN, which adopts a SINN architecture previously validated for frequency analysis and combines it with an autoencoder-based network through multi-task learning. SINN employs both supervised learning and unsupervised observation consistency, while the autoencoder estimates source-presence probabilities. Their complementary outputs are jointly optimized and fused to produce the final DOA spectrum. Experimental results demonstrate improved robustness under varying acoustic conditions.

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Acknowledgment: This work was supported by the Korea Research Institute for Defense Technology Planning and Advancement (KRIT) grant funded by the Korea government (Defense Acquisition Program Administration (DAPA)) (No. KRIT-CT-23-026, Integrated Underwater Surveillance Research Center for Adapting Future Technologies, 2026)

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Shallow-Water Measurement of Underwater Radiated Noise from a Small-Sized USV for Developing an Accredited Underwater Noise Assessment System

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Keywords: Underwater radiated noise (URN), Unmanned marine vehicle (UMV), Unmanned surface vehicle (USV), Source level, Shallow-water environment

Abstract: With the recent active development and deployment of unmanned marine vehicles, interest in the underwater radiated noise they generate has also been increasing. However, an accredited certification system for measuring and evaluating the underwater radiated noise of marine unmanned vehicles has yet to be fully established worldwide. Although ISO 17208-3 provides the latest standard for evaluating surface vehicles' underwater radiated noise in shallow-water environments, the relatively low noise levels of small-sized unmanned surface vehicles (USVs) often prevent achieving a sufficient signal-to-noise ratio relative to ambient noise, so the measurement conditions specified in the standard may need refinement for their application. As a

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preliminary step toward establishing an accredited certification system, the present study aims to identify conditions under which underwater radiated noise can be accurately measured in shallow-water environments, based on field measurements conducted with a small-sized commercial USV. During the measurements, the USV was operated at two different speeds, and its distance from the moored hydrophone deployment point was varied over several values. By analyzing the noise signals received by the underwater hydrophone during the USV's operating periods, we determined appropriate hydrophone stand-off distances for underwater radiated noise measurements of small-sized USVs and a suitable data window length for noise analysis. These results can provide empirical supporting data for establishing the evaluation procedures for the underwater radiated noise of marine unmanned vehicles.

Acknowledgment: This research was supported by the Korea Institute of Marine Science & Technology Promotion (KIMST) funded by the Ministry of Oceans and Fisheries (RS-2023-00256122)

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Dechannelized Representation for Passive Sonar Target Classification

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Keywords: underwater acoustic target recognition; DeepShip dataset; passive sonar; domain shift

Abstract: Underwater acoustic target recognition (UATR) often suffers from weak generalization because propagation-induced channel variation distorts the received signal. This paper proposes a dechannelization representation that reduces channel-dominant variation and emphasizes source-related spectral structure. Starting from a source–channel observation model, the proposed method trend–residual view of the time-varying propagation channel. A temporal moving average is used to estimate a channel-dominant trend in the received spectrum, and the residual is squared and averaged over a longer time scale to obtain a stable, source-dominant representation. Bellhop-based simulations demonstrate that the proposed representation becomes more source-consistent than the raw received spectrum. Experiments on the DeepShip dataset under progressively challenging split scenarios, especially location-mismatched conditions, show that the proposed representation consistently improves classification robustness over the conventional STFT baseline

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Acknowledgment: This work was supported by the Korea Research Institute for Defense Technology Planning and Advancement (KRIT) grant funded by the Korea government (Defense Acquisition Program Administration (DAPA)) (No. KRIT-CT-23-026, Integrated Underwater Surveillance Research Center for Adapting Future Technologies, 2026)

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Passive Acoustic Monitoring of Polar Soundscapes at KOPRI

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Keywords: Polar soundscapes, Passive acoustic monitoring, Marine mammal, Machine learning

Abstract: The Korea Polar Research Institute (KOPRI) conducts active and passive acoustic research in the Arctic and Antarctic Oceans to understand how climate change and increasing human activities are altering polar marine ecosystems. In this presentation, we introduce KOPRI's passive acoustic monitoring (PAM) research and its contributions to understanding polar ocean soundscapes. KOPRI focuses on characterizing long-term variability in soundscapes and ambient noise, assessing anthropogenic noise and its ecological implications, monitoring the occurrence and acoustic behavior of polar marine mammals, and developing automated analytical and sound-

propagation methods for large acoustic datasets. Long-term PAM deployments in the East Siberian Sea and Ross Sea show that underwater noise can rise substantially when anthropogenic sources, particularly seismic airguns, coincide with reduced sea ice. Analyses of bearded and leopard seal calls have supported assessments of species occurrence and acoustic behavior and identified previously undescribed calls, including a triple ascending trill. Convolutional neural networks have been developed to classify seal calls, airgun pulses, vessel noise, and other acoustic events in continuous recordings. Grad-CAM and a feature-based framework incorporating biologically and acoustically relevant features have been used to interpret and ecologically validate model decisions. Power spectral density and warping-transform analyses of broadband airgun signals across the East Siberian continental shelf have also quantified transmission loss and normal-mode dispersion and estimated water depth and seabed sound speed and density. Future research will extend this framework to the quantitative assessment of ship-radiated noise in the Arctic Ocean, including source-level estimation, vessel-specific acoustic characterization, propagation modeling under changing sea-ice and oceanographic conditions, and evaluation of potential noise exposure footprints for Arctic marine organisms.

Acknowledgment: This research was supported by a grant from the Korea Institute of Marine Science & Technology Promotion (KIMST), funded by the Ministry of Oceans and Fisheries (RS-2021-KS211500 and RS-2022-KS221661).

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Temporal and Spatial Variability in Underwater Soundscapes Around the Japanese Coast

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Keywords: Underwater acoustics, Fish chorus, Ocean ambient noise, Offshore wind farm

Abstract: Underwater noise has received increasing attention in recent decades. Assessing its effects on aquatic animals requires characterization of both source spectrum and ambient noise conditions, which largely determine sound detection range. Regional and seasonal variability in underwater soundscapes around the Japanese coast were examined. Autonomous underwater recorders were deployed at ten coastal sites for continuous 15-day observations during summer and winter to characterize ambient sound levels. The results revealed marked seasonal and diel differences: in summer, evening fish choruses were detected at most sites, whereas in winter, high-intensity, low-frequency broadband noise—primarily driven by strong northwest monsoons—dominated, particularly along the Japan Sea coast. These patterns underscore the dynamic interplay of biophony and geophony in Japanese coastal waters. For the same anthropogenic noise source level, the maximum effective range varies seasonally, becoming longer in summer and shorter in

winter due to differences in masking level by ambient noise. Our findings emphasize the importance of jointly considering source and ambient noise levels in underwater-noise impact assessments. Appropriate survey planning requires careful site selection based on local conditions.

Acknowledgment: This study was supported by the Ministry of the Environment of Japan under the 2020 Underwater Sound Monitoring Project.

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Distribution and density of Fish Resource in Marine Aggregate Extraction Areas Using Acoustic Surveys and Environmental DNA Analysis

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Keywords: Hydro-acoustic survey, aggregate extraction, Exclusive Economic Zone (EEZ), Environmental DNA (eDNA)

Abstract: Sand and gravel resources within exclusive economic zones (EEZs) are widely used as construction materials worldwide. However, marine aggregate extraction can adversely affect marine ecosystems by altering seabed topography and removing bottom sediments. This study evaluated changes in fish distribution associated with marine aggregate extraction through seasonal acoustic surveys and environmental DNA (eDNA) analyses conducted from 2021 to 2023 at extraction and reference sites in the coastal Yellow Sea, the Yellow Sea EEZ, and the EEZ in the South Sea of Korea. Fish distribution and acoustic density were assessed using 38 and 120 kHz scientific echosounders, while species diversity was characterized through eDNA analysis. In the coastal Yellow Sea, the mean nasc was relatively high at the extraction sites in 2021 but remained low at both the extraction and reference sites in 2022 and 2023. In the Yellow Sea EEZ, the mean

NASC was higher at the extraction sites than at the reference sites in both 2022 and 2023, with high acoustic backscatter repeatedly observed along Transects 6 and 7. Within the EEZ in the South Sea of Korea, the sites exhibiting higher NASC values varied between the extraction and reference areas depending on the year and season. Fish aggregations occurred either throughout the water column or locally within specific depth ranges. Overall, fish distribution in marine aggregate extraction areas exhibited substantial spatial and temporal variability rather than a consistent pattern attributable solely to extraction activities. These findings highlight the need for long-term, repeated monitoring that integrates acoustic surveys with eDNA analysis to accurately assess the effects of marine aggregate extraction.

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Integrated Active and Passive Acoustic Monitoring of Finless Porpoises and Fish Aggregations at a Set-net Fishing Ground in Goseong, South Korea

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Keywords: finless porpoise, passive acoustic monitoring, active acoustics, marine protected area, set-net fishery

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Abstract: The narrow-ridged finless porpoise (*Neophocaena asiaeorientalis*) is one of the most common small cetaceans inhabiting Korean coastal waters, yet its fine-scale association with prey distribution and coastal fisheries remains poorly understood. To address this knowledge gap, an integrated active and passive acoustic monitoring system was established at a set-net fishing ground in Samsan-myeon, Goseong, South Korea, approximately 9 km from the Goseong Marine Protected Area. A Wideband Autonomous Transceiver (WBAT) echosounder was deployed along the leader net of the set-net, whereas an F-POD passive acoustic recorder was installed at the terminal end of the trap. Operating simultaneously at the same fishing ground, the two systems continuously monitored fish aggregations, underwater scattering layers, and finless porpoise echolocation clicks from 14 June to 30 October 2026. This integrated acoustic system enables concurrent observations of predator occurrence and prey-field dynamics without disturbing the animals or fishing operations. The monitoring program aims to quantify temporal associations between finless porpoise occurrence and fish aggregations, evaluate the influence of set-net fishing grounds on finless porpoise habitat use, and establish a long-term baseline for ecosystem monitoring in Korean coastal waters. The integrated observations are expected to improve understanding of predator–prey interactions and provide scientific information to support ecosystem-based fisheries management and conservation of finless porpoises.

Acknowledgment: The authors sincerely thank Mr. Hee-Sung Hwang of Sanho Fisheries (Goseong, Republic of Korea) for generously providing access to the set-net fishing ground and supporting the field deployment of the acoustic monitoring system. This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT)(RS-2025-00520657).

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Frequency-Dependent Characteristics of Underwater Ambient Noise at Different hydrophone Depths in the Deep East Sea of Korea

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Tuesday 13 - 09:50 ~ 11:40

Keywords: Korea East Sea, Underwater Sound, Ocean Noise, Ship noise, Deep-water

Abstract: The East Sea of Korea has a deep basin and steep bathymetry, which can produce complex sound propagation in deep water. Noise generated near the sea surface by wind and ships may reach greater depths or become locally stronger, depending on frequency. At a deep-water station with a bottom depth of approximately 1,920 m, underwater background noise was analyzed at receiver depths of 100, 150, 200, 300, 400 m. At all depths, noise levels decreased with increasing frequency. Noise levels also generally decreased with receiver depth, with relatively high levels at 100 m and low levels at 400 m. However, in the 1 ~ 5 kHz band the decrease with depth was not consistent. The receivers at 200 m, near the sound channel and at 300 m showed slightly higher levels than nearby depths at some frequencies. This suggests that noise generated near the sea surface does not simply weaken with depth, but changes with frequency and sound propagation conditions. Therefore, both receiver depth and frequency band should be considered when interpreting underwater background noise in the deep East Sea. Future correlation analyses with ship noise and wind speed will help identify the main factors affecting noise variation across frequencies.

Acknowledgment: This research was financially supported under the “Development of Maritime Domain Awareness Technology for Sea Power Enhancement (PEA0421)” project funded by the Korea Institute of Ocean Science & Technology.

ACOUSTICAL ASSESSMENT Tuesday 13 - 09:50 ~ 11:40

Environmental Controls on Low-Frequency Underwater Sound Levels under Changing Sea-Ice Conditions in the East Siberian Arctic Shelf

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ACOUSTICAL ASSESSMENT Tuesday 13 - 09:50 ~ 11:40

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Keywords: Arctic underwater noise, Passive acoustic monitoring, Anthrophony, Sea ice, Environmental controls

Abstract: Rapid changes in Arctic sea-ice conditions are increasing accessibility to the Arctic Ocean and the potential for human activities, raising concerns about how environmental variability and anthropogenic activities influence underwater soundscapes. This study examines the relationships between environmental conditions and low-frequency underwater sound levels on the East Siberian Arctic Shelf using acoustic and environmental datasets collected from 2017 to 2024. Third-octave-band sound pressure levels at 63 and 125 Hz, which are commonly used as low-frequency indicators associated with anthropogenic noise, were analyzed in relation to sea-ice concentration, sea-ice drift, sea surface temperature, 2 m air temperature, and wind speed. Pearson correlation analysis and principal component analysis were applied to quantify pairwise relationships and identify dominant patterns of covariability among acoustic and environmental variables. To assess the role of sea ice as a modulating factor, the dataset was classified into high-, intermediate-, and low-ice regimes. The results show that the relationships between underwater sound levels and environmental variables vary across sea-ice regimes. These findings suggest that sea-ice conditions modulate the environmental controls on low-frequency underwater sound levels and should be considered when interpreting anthropogenic noise indicators in the Arctic marine environment.

Acknowledgment: This research was supported by the Korea Polar Research Institute (KOPRI, PE26210) and supported by the Korea Institute of Marine Science & Technology Promotion (KIMST) grant funded by the Ministry of Oceans and Fisheries (KIMST RS-2021-KS211500).

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